

Recalibrating EuroLeague Performance Index Rating for outcome-concordant team ledgers: a season-locked historical evaluation

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ABSTRACT

Transparent performance indices are useful when their arithmetic can be reconstructed from official records, but equal unit weights across heterogeneous box-score events need not produce a team ledger that is concordant with the completed-game outcome. We introduce RMS-matched Performance Index Rating (RM-PIR), a constrained, season-locked recalibration of the EuroLeague Performance Index Rating that fixes points at coefficient one, estimates bounded relative magnitudes for nonpoint actions using only completed prior seasons, and subsequently matches the calibration-window root-mean-square scale of the official nonpoint winner–loser contrasts. We evaluated the method in a retrospective five-target rolling-origin backtest covering 1,690 target games from 2021–22 through 2025–26, with an additional 2020–21 through 2025–26 population of 2,018 completed games used for data-integrity and descriptive analyses. Across the five target seasons, complete-score discordances decreased from 150 under official PIR to 104 under RM-PIR; nonpoint discordances decreased from 293 to 209. In 2025–26, complete-score discordances decreased from 27 to 21 games, while nonpoint discordances decreased from 68 to 45. The complete-score effect was more modest and sensitive to dependence-aware uncertainty and tie-equivalence choices, whereas the nonpoint mechanism change was larger and more consistent across the reported sensitivities. RMS matching did not determine the sign of nonpoint ordering changes. The study is explicitly an outcome-concordance analysis of a transparent team ledger: it does not establish causal player value, predictive superiority, or context-complete individual impact. The method provides a reproducible framework for prospective evaluation using a specification frozen before an untouched future season.

Introduction

Professional basketball is simultaneously an elite competition, a labour market, a media property and a large-scale entertainment product. The economic setting in which EuroLeague performance is evaluated is therefore substantial. A 2026 valuation prepared by JB Capital and released by Euroleague Basketball estimated the combined enterprise value of the competition and its licensed clubs at more than EUR 3.2 billion, comprising EUR 1.41 billion for the league and approximately EUR 1.8 billion for licensed clubs¹. Enterprise value is not equivalent to annual revenue, but it indicates the scale of the assets, contracts and strategic decisions associated with the competition. The link between commercial capacity and sporting expenditure is also embedded in league governance: the EuroLeague Competitive Balance Standards connect club revenues to player-remuneration thresholds and establish a base net remuneration level of EUR 10 million for the 2025–26 season^{5,4}. In this environment, the quality of performance measurement is not merely a statistical concern; it also shapes how clubs interpret the sporting return from high-value roster investments.

Clubs allocate finite resources across player recruitment, contract renewal, transfers and buyouts, coaching, medical support, player development and analytics. Quantitative analysis is used in professional basketball to evaluate player and team performance and to inform tactical and personnel decisions^{3,4}, while one NBA study associated organisational investment in analytics with improved team-level performance⁵. EuroLeague-specific research has used objective player-performance profiles to draw practical implications for player management⁶; in the Spanish ACB, PIR has also been used as the performance input to roster-selection optimisation⁷. Clubs' proprietary decision systems are not publicly observable, and no responsible investment decision should be reduced to one statistic. Nevertheless, a transparent performance index provides a common descriptive layer for post-game review, longitudinal monitoring, comparison of players in similar roles and evaluation of the team as a whole. Because the Performance Index Rating (PIR) is additive, the credits and penalties assigned to player actions are aggregated in a team ledger. This structure makes PIR especially attractive for examining both individual production and the realised collective output of a roster, provided that the underlying action weights are defensible.

The same index also affects the public interpretation of EuroLeague basketball. Euroleague Basketball reported that the 2025–26 regular season generated a cumulative audience of 508.17 million across live, delayed and highlights formats, 1.49

billion video views on league-owned social-media platforms and a record 3,251,706 in-arena spectators⁸. At this scale, statistics are part of the entertainment product: they organise post-game discussion, support comparisons, create historical records and give fans a concise account of why a performance mattered. PIR has direct institutional salience because the EuroLeague MVP of the Round is awarded to the player with the highest PIR among players whose teams won⁹. The explicit winning-team condition is important: it already recognises that individual box-score production must be interpreted in relation to the team outcome.

Performance indicators compress multidimensional sporting activity into quantities that can be compared across games, teams and seasons. Their usefulness therefore depends on a clearly defined purpose, an appropriate basis for comparison and an explicit account of what is omitted¹⁰. Longitudinal or cross-competition comparison also requires attention to differences in rules and recorded-statistic distributions¹¹. Media reporting necessarily compresses a complex game into a limited number of accessible narratives, and a single index is valuable precisely because it can be communicated immediately and reconstructed from an official box score. Its limitation is a consequence of treating a multidimensional performance as one scalar quantity. Conventional box-score indices do not directly observe off-ball spacing, screen creation, defensive positioning, matchup difficulty, shot deterrence, lineup interaction or the leverage of an action at a particular game state; lineup-, possession- and tracking-based methods explicitly model some of these contextual or spatial structures^{12–16}. A reporting gap can arise when a partial index is read as a complete estimate of player impact. The gap is more consequential when the index's arithmetic assigns a larger team total to the losing team, because the same formula then distributes player credit in a way that is less consistent with the completed-game result. A more outcome-concordant index cannot eliminate the need for video, lineup and contextual analysis, but it can reduce this inconsistency at the source.

Official EuroLeague PIR has a major practical advantage: it is transparent and can be reconstructed from the competition's standardised team and player box-score fields^{17,18}. It awards one unit for points, rebounds, assists, steals, blocks and fouls drawn, and subtracts one unit for missed field goals, missed free throws, turnovers, blocks against and fouls committed; the complete formula is given in Equation 1. Missed field goals and missed free throws are derived from attempts and makes. This convention is easy to audit, but it implicitly treats heterogeneous events as equivalent units of credit or cost. The broader basketball-analytics literature has long emphasised possession-based comparison and the need to relate recorded actions to team success^{19,20}. EuroLeague studies have found that shooting efficiency, rebounding, assists, steals, turnovers and fouling are not equally associated with winning, and that their empirical importance can vary with competitive stage and game context^{21–23}. Related evidence from other professional basketball competitions likewise shows heterogeneous associations between recorded actions and team success^{24–26}. Equal unit weights are therefore an auditable convention rather than an empirically calibrated statement that every recorded action contributes equally to the completed-game ledger.

For an additive post-game index, we selected the proportion of completed games in which the winning team's total exceeds the losing team's total as the team-level estimand. This is a normative outcome-concordance criterion, not an objective error definition: PIR can also be read as box-score production without requiring it to reproduce the winner. Nor does the criterion imply that every valuable individual performance must occur in a victory or that a box-score rating measures a player's causal effect. Because points define the winner, the complete-score criterion is mechanically related to the outcome. A points-only score would be perfectly concordant while discarding rebounds, assists, steals, blocks, fouls, missed shots and turnovers. The nonpoint analysis is therefore the key mechanism check: it tests the learned relative action direction apart from the direct point-margin contribution.

Winner-concordance is treated here as a coherence property of a completed-game descriptive team ledger: under the selected criterion, a summary of the two teams' recorded actions should not assign the larger total to the losing team once the game is complete. It is not proposed as a universal definition of basketball performance, team quality or individual value. The nonpoint objective has a distinct purpose. Points are excluded from estimation because their outcome-defining margin makes a points-only construction mathematically sufficient for perfect winner ordering; conditional on retaining the other official actions, winner alignment of their combined nonpoint contrast is the criterion chosen to reallocate their relative credits and penalties.

Existing EuroLeague studies have identified box-score variables associated with winning or season success^{18,21–23}. Outcome-oriented reweighting also predates the present study. Celeux and Robert estimated an objective team-efficiency rate from season wins, evaluated it against regular-season rankings and then considered a player-oriented form⁵¹. Marmarinos et al. developed a Total Performance Index for EuroLeague players and evaluated its structural validity and reliability in comparison with PIR⁵²; a later Spanish-language working paper proposed a min–max transformation of PIR for player comparison⁵³. Berri linked NBA player statistics to team wins econometrically²⁷, and WINSORE Revisited derives box-score weights endogenously using NBA and EuroLeague data and evaluates the resulting measures against PIR²⁸. Other recent approaches address different estimands and structures, including hierarchical NBA athlete and team ratings²⁹ and CBA game-outcome prediction and explainability^{26,30}.

These precedents establish that data-derived basketball weighting, outcome-oriented efficiency scoring and PIR modification

Table 1. Notation, score components and conventions. Favourable actions enter with a positive sign and unfavourable actions with a negative sign.

Label	Definition	Use in this study
PIR	Performance Index Rating	Official additive EuroLeague box-score index.
RM-PIR	RMS-matched PIR	Team-calibrated, season-locked recalibration of the same action ledger.
PTS	Points	Fixed coefficient one; excluded numerically from fitting and tuning.
TR, AST, STL	Total rebounds, assists, steals	Favourable non-PTS components.
BLK, FD	Blocks, fouls drawn	Favourable components constrained to share raw magnitudes with BLKA and FC, respectively.
MFG, MFT	Missed field goals, missed free throws	Unfavourable components, derived as attempts minus makes.
TO, BLKA, FC	Turnovers, blocks against, fouls committed	Unfavourable non-PTS components.
$\hat{\mathbf{a}}_Y$	Raw action-group magnitudes	Eight pre-rescaling magnitudes fitted from the calibration window for target season Y and constrained to $[0.5, 1.5]$.
γ	RMS scale factor	Matches the calibration-sample RMS of official and rescaled fitted winner–loser non-PTS contrasts for target season Y .
$\Delta_{S,g}$	Winner-minus-loser score difference	Concordant if $\Delta_{S,g} > 10^{-9}$; numerical ties and reversals are reported separately.
W/T/L	Winner higher/tie/loser higher	Complete partition of target-season game ordering.
Rank change	Official rank minus RM-PIR rank	Positive values denote upward movement under RM-PIR.
SRE	Symmetric relative error	$200 a - b /(a + b)$; unsigned and secondary to the signed percentage-point change.

are not new. The unresolved problem addressed here is narrower: whether the official EuroLeague PIR ledger can be reweighted so that its team totals are more concordant with completed-game outcomes, without collapsing to the degenerate points-only solution or abandoning the official fields, signs and additive interpretation. The intended scope is therefore a competition-specific team-ledger recalibration and its exact downstream player decomposition, not a replacement for contextual player-impact models. The novelty claimed is the specified combination of preserved official fields and signs, a point coefficient fixed at one, bounded nonpoint estimation from completed prior seasons, RMS anchoring, season locking, game-level concordance assessment and explicit team-only residual accounting.

We therefore evaluate RMS-matched PIR (RM-PIR), a competition-specific recalibration of the official action ledger. Points remain fixed at coefficient one; bounded nonpoint weights are estimated from winner–loser differences in completed prior-season team box scores; and a positive factor matches the root mean square of fitted and official nonpoint contrasts within the calibration window. The primary operational outcome is complete-score team concordance, while nonpoint concordance tests the fitted action direction independently of the point margin. The principal evaluation is a five-target retrospective rolling-origin backtest, with each coefficient vector locked before its target season. Player analyses are restricted to exact algebraic decomposition and secondary descriptive diagnostics. The paper asks whether this constrained recalibration improves conditional historical team-ledger concordance, how stable that conclusion is across seasons and sensitivity analyses, and what reallocation follows when the same locked ledger is decomposed to player rows. It does not propose a causal player-value construct, individual validation or prospective prediction.

Methods

Official EuroLeague PIR: the existing additive ledger

Official PIR is an additive box-score ledger reported in EuroLeague statistics³¹. Under the unit-weight convention described in EuroLeague performance research¹⁸, points (PTS), total rebounds (TR), assists (AST), steals (STL), blocks (BLK) and fouls drawn (FD) each receive unit positive credit, whereas missed field goals (MFG), missed free throws (MFT), turnovers (TO), blocks against (BLKA) and fouls committed (FC) each receive unit negative credit. Points therefore enter with coefficient +1, as do the other favourable actions, while the unfavourable actions enter with coefficient -1. The same formula can be evaluated on an official player row or an official team row. For any such row u in game g , the ledger is

$$\begin{aligned} \text{PIR}_{ug} = & \text{PTS}_{ug} + \text{TR}_{ug} + \text{AST}_{ug} + \text{STL}_{ug} + \text{BLK}_{ug} + \text{FD}_{ug} \\ & - \text{MFG}_{ug} - \text{MFT}_{ug} - \text{TO}_{ug} - \text{BLKA}_{ug} - \text{FC}_{ug}. \end{aligned} \quad (1)$$

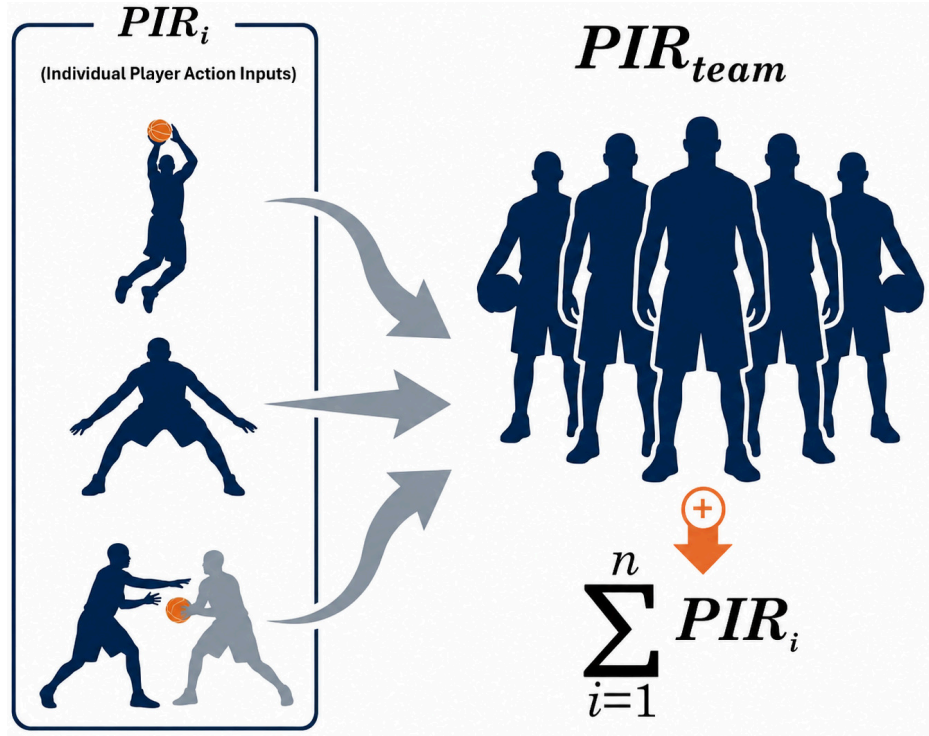


Figure 1. Construction of the existing official EuroLeague Performance Index Rating (PIR) at player and team levels. Recorded box-score actions are evaluated under the official PIR formula to obtain each player's PIR_i . Summation across the n participating players yields the player-attributed component of official team PIR. The schematic represents official PIR only; it does not depict RM-PIR calibration. Official team rows may additionally include actions that are not allocated to player rows, which are retained explicitly through the team-only residual defined later in the Methods.

The two missed-shot components were derived from the corresponding attempt and make fields:

$$MFG_{ug} = FGA_{ug} - FGM_{ug}, \quad MFT_{ug} = FTA_{ug} - FTM_{ug}, \quad (2)$$

where FGA and FGM denote field-goal attempts and makes, respectively, and FTA and FTM denote free-throw attempts and makes.

The formula preserves a transparent mapping from recorded box-score actions to the reported index. MFG and MFT are not separate makes-based inputs; they are derived immediately from attempts and makes as shown above. Because the same row formula applies at player and team level, player-attributed values can be summed toward the team ledger, while any official team-row actions not allocated to player rows can be retained explicitly rather than silently reassigned. Figure 1 illustrates this existing official system only.

The principal advantage of this construction for the present study is its transparency and reconstructability. Its corresponding limitation is that the same absolute unit magnitude is assigned to heterogeneous recorded actions. The remainder of the method asks whether those nonpoint magnitudes can be recalibrated while preserving the official ledger structure.

Outcome-concordance criterion and the complete-score fitting gap

The team-level criterion examined here is winner–loser ordering of the completed-game ledger. For a score S in game g , let $\Delta_{S,g} = S_{W_g} - S_{L_g}$, where W_g and L_g denote the winning and losing teams. Strict concordance is defined by $\Delta_{S,g} > 10^{-9}$; numerical ties and reversals are retained as separate categories, and both are discordant under the strict analysis rule. This is an author-selected outcome-concordance criterion for a completed-game team ledger, not an objective definition of player value. Complete-score concordance is the primary operational outcome because it describes the deployed team score. Nonpoint concordance is the mechanism outcome because it evaluates the relative action direction after removing the point margin that determines the winner.

To express the official ledger and the winner–loser comparison compactly, collect the ten non-PTS fields and their official favourable/unfavourable directions as follows.

Let u denote an official player or team row in game g . Collect the ten non-PTS fields in the unsigned action-count vector

$$\mathbf{X}_{ug} = (\text{TR}_{ug}, \text{AST}_{ug}, \text{STL}_{ug}, \text{BLK}_{ug}, \text{FD}_{ug}, \text{MFG}_{ug}, \text{MFT}_{ug}, \text{TO}_{ug}, \text{BLKA}_{ug}, \text{FC}_{ug})^T \in \mathbb{R}^{10}, \quad (3)$$

whose j th entry is denoted by X_{ugj} , and define the official sign vector

$$\mathbf{s} = (1, 1, 1, 1, 1, -1, -1, -1, -1, -1)^T. \quad (4)$$

Equation 1 is therefore equivalent to

$$\text{PIR}_{ug} = \text{PTS}_{ug} + \mathbf{s}^T \mathbf{X}_{ug}. \quad (5)$$

For completed game g , define the unsigned action difference and the strictly positive point margin by

$$\mathbf{d}_g = \mathbf{X}_{Wg,g} - \mathbf{X}_{Lg,g}, \quad \Delta_g^{\text{PTS}} = \text{PTS}_{Wg,g} - \text{PTS}_{Lg,g} > 0. \quad (6)$$

The signed winner–loser design vector is obtained by applying the official signs componentwise:

$$\begin{aligned} \mathbf{x}_g &= \mathbf{s} \odot \mathbf{d}_g \\ &= (\Delta \text{TR}_g, \Delta \text{AST}_g, \Delta \text{STL}_g, \Delta \text{BLK}_g, \Delta \text{FD}_g, -\Delta \text{MFG}_g, -\Delta \text{MFT}_g, \\ &\quad -\Delta \text{TO}_g, -\Delta \text{BLKA}_g, -\Delta \text{FC}_g)^T, \end{aligned} \quad (7)$$

where $\odot$ denotes elementwise multiplication. Winner identity determines the orientation, but neither the numerical point total nor the winning margin enters $\mathbf{x}_g$. The official non-PTS and complete PIR contrasts follow directly:

$$\Delta_g^{\text{PIR,nonPTS}} = \mathbf{x}_g^T \mathbf{1}_{10}, \quad \Delta_g^{\text{PIR}} = \Delta_g^{\text{PTS}} + \mathbf{x}_g^T \mathbf{1}_{10}. \quad (8)$$

Although the point margin is strictly positive, the signed nonpoint term can be negative enough for the complete official PIR contrast to be tied or reversed under the study criterion.

More generally, an additive score with signed non-PTS coefficient vector $\mathbf{c}$ has row score and winner–loser difference

$$S_{ug}(\mathbf{c}) = \text{PTS}_{ug} + \mathbf{X}_{ug}^T \mathbf{c}, \quad \Delta_g(\mathbf{c}) = \Delta_g^{\text{PTS}} + \mathbf{d}_g^T \mathbf{c}. \quad (9)$$

Setting $\mathbf{c} = \mathbf{0}$ gives

$$\Delta_g(\mathbf{0}) = \Delta_g^{\text{PTS}} > 0, \quad (10)$$

and hence perfect winner ordering.

The last identity exposes the central fitting problem. Because the winning team necessarily has a positive point margin, an unrestricted procedure allowed to eliminate the nonpoint ledger can obtain perfect winner ordering simply by setting every nonpoint coefficient to zero. That solution would satisfy the complete-score ordering criterion while discarding rebounds, assists, steals, blocks, fouls, missed shots and turnovers, and would therefore defeat the purpose of retaining a meaningful PIR-style action ledger.

These facts define the design requirements for the alternative considered here. It must retain the official PIR fields and their favourable or unfavourable directions; keep points fixed at coefficient one; preserve an additive, reconstructable ledger; prevent the trivial points-only solution; estimate relative nonpoint action magnitudes from completed prior data; remain interpretable on a scale comparable with official PIR; and lock every quantity before the target season so that target observations do not enter fitting, selection or scale matching.

From official PIR to RMS-matched PIR

RMS-matched Performance Index Rating (RM-PIR) is the response to those requirements. In plain terms, the procedure leaves points unchanged, isolates the nonpoint portion of the official ledger, forms winner–loser contrasts from those nonpoint actions, estimates bounded relative magnitudes while retaining the official favourable/unfavourable signs, and then post-scales the fitted nonpoint direction to the root-mean-square (RMS) scale of official nonpoint contrasts in the same calibration window. The resulting coefficients are locked before the next target season. The complete RM-PIR score is then reconstructed by adding the fixed point contribution back to the recalibrated nonpoint ledger. Thus the fitted object is the nonpoint action direction, whereas the primary operational evaluation is the complete score after the point margin has been restored.

The notation and scoring conventions used in the construction are summarised in Table 1. Figure 2 then shows the annual flow from completed prior-season team box scores to a locked target-season score.

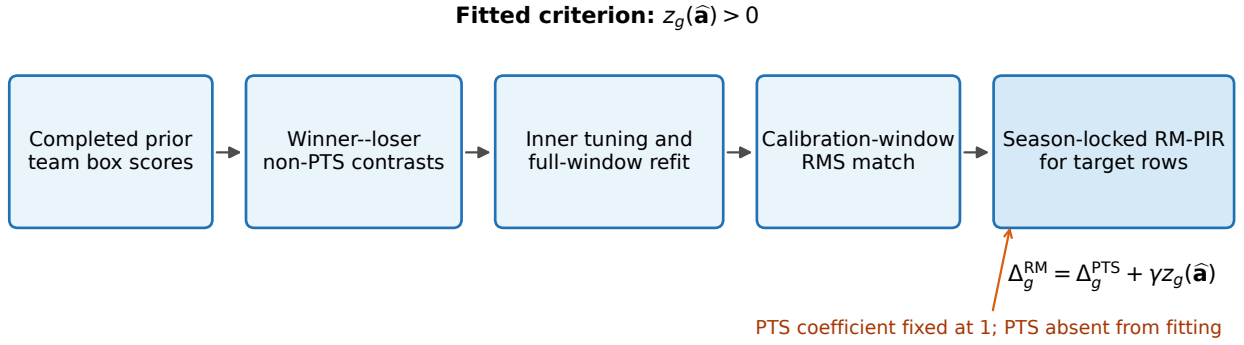


Figure 2. Nested, season-locked RM-PIR calibration. Completed prior-season team box scores determine winner–loser non-PTS contrasts. Hyperparameters are selected within the eligible calibration window; the selected model is refitted on the complete window; and a common RMS factor places the fitted non-PTS contrast on the official calibration-sample scale. Points remain fixed at coefficient one and enter only when the complete target-season score is assembled.

RM-PIR action-weight parameterisation

The next step is to allow the official nonpoint fields to differ in relative magnitude without changing which actions are credited and which are penalised. RM-PIR therefore estimates positive raw magnitudes and applies the official sign vector separately.

The restrictions below are substantive safeguards on the meaning, scale and temporal relevance of the recalibrated ledger. Points retain their official coefficient of one and are excluded from estimation so that the outcome-defining point margin cannot determine the fitted nonpoint weights. The official favourable and unfavourable signs are fixed because recalibration is intended to change the magnitude, not the meaning, of each recorded action. Raw nonpoint magnitudes are bounded within $[0.5, 1.5]$ around the official unit convention to prevent elimination of the retained action ledger and to limit departure from the parent index while permitting relative reallocation; wider bounds are examined as sensitivities. BLK and BLKA, and FD and FC, share magnitudes because each pair largely records opposing sides of the same event class and therefore generates strongly redundant signed contrasts. This rationale is supported by the 2022–23 through 2024–25 calibration-window diagnostic, for which grouping reduced the maximum standardised condition index from 100.79 to 3.44. The use of at most three recent completed seasons is an author-selected compromise between sampling support and temporal relevance under changing rosters, competition composition and potentially evolving recording conventions; shorter windows are examined separately. RMS is the primary anchor because it acts directly on the oriented nonpoint contrasts being modelled, matches the fitted contrasts’ uncentred second moment to that of the official contrasts through a unique positive scale factor whenever the fitted RMS is nonzero, and leaves their zero boundary unchanged. It is not assumed to be the uniquely correct scale convention, and alternative anchors are included in the sensitivity analyses.

RM-PIR estimates eight positive raw group magnitudes,

$$\mathbf{a} = (a_{\text{TR}}, a_{\text{AST}}, a_{\text{STL}}, a_{\text{B}}, a_{\text{F}}, a_{\text{MFG}}, a_{\text{MFT}}, a_{\text{TO}})^{\text{T}} \in [0.5, 1.5]^8. \quad (11)$$

The expansion matrix $\mathbf{E} \in \{0, 1\}^{10 \times 8}$ implements the stated BLK–BLKA and FD–FC equality restrictions. In the component order of Equation 3,

$$\mathbf{E}\mathbf{a} = (a_{\text{TR}}, a_{\text{AST}}, a_{\text{STL}}, a_{\text{B}}, a_{\text{F}}, a_{\text{MFG}}, a_{\text{MFT}}, a_{\text{TO}}, a_{\text{B}}, a_{\text{F}})^{\text{T}}. \quad (12)$$

Thus BLK and BLKA share one unsigned raw magnitude, as do FD and FC, while their favourable and unfavourable directions remain encoded by $\mathbf{s}$. The fitted non-PTS contrast is

$$z_g(\mathbf{a}) = \mathbf{x}_g^{\text{T}} \mathbf{E}\mathbf{a}. \quad (13)$$

Because $\mathbf{E}\mathbf{1}_8 = \mathbf{1}_{10}$, the official non-PTS contrast in Equation 8 is nested at $\mathbf{a} = \mathbf{1}_8$. The primary calibration target is $z_g(\mathbf{a}) > 0$, which requires the non-PTS action ledger itself to favour the winner.

Estimation and chronological model selection

With the fitted object defined, estimation can be stated directly. The discontinuous count of nonpoint discordances is replaced by a smooth decreasing surrogate, while a ridge term shrinks the expanded magnitudes toward the official unit-weight convention. The raw magnitudes remain bounded away from zero, so the fitted nonpoint ledger cannot collapse to the points-only solution.

The discontinuous indicator $\mathbf{1}\{z_g \leq 0\}$ is unsuitable for gradient-based estimation. For transition width $\tau > 0$, RM-PIR uses the decreasing differentiable surrogate

$$\ell_\tau(z) = \log \left[1 + \exp \left(-\frac{z}{\tau} \right) \right], \quad \frac{d\ell_\tau(z)}{dz} = -\frac{1}{\tau \{1 + \exp(z/\tau)\}}. \quad (14)$$

For a calibration set $\mathcal{A}$ containing $n_{\mathcal{A}}$ games, define

$$\mathcal{L}_{\mathcal{A}}(\mathbf{a}; \tau, \alpha) = \frac{1}{n_{\mathcal{A}}} \sum_{g \in \mathcal{A}} \ell_\tau(z_g(\mathbf{a})) + \frac{\alpha}{10} \|\mathbf{E}\mathbf{a} - \mathbf{1}_{10}\|_2^2. \quad (15)$$

The bounded raw estimator is

$$\hat{\mathbf{a}}_{\mathcal{A}}(\tau, \alpha) = \arg \min_{\mathbf{a} \in [0.5, 1.5]^8} \mathcal{L}_{\mathcal{A}}(\mathbf{a}; \tau, \alpha). \quad (16)$$

The first term is a smooth surrogate for non-PTS discordance. The second averages the squared deviation of the ten expanded field magnitudes from the official unit-weight convention and stabilises correlated action contrasts. The denominator is 10 rather than 8 because BLK and BLKA, and FD and FC, remain separate scored fields even though each pair shares a fitted magnitude; the penalty therefore weights each ledger field once. Applying the chain rule to Equation 14 gives

$$\begin{aligned} \nabla_{\mathbf{a}} \mathcal{L}_{\mathcal{A}} &= -\frac{1}{n_{\mathcal{A}} \tau} \sum_{g \in \mathcal{A}} \frac{\mathbf{E}^\top \mathbf{x}_g}{1 + \exp\{z_g(\mathbf{a})/\tau\}} \\ &\quad + \frac{2\alpha}{10} \mathbf{E}^\top (\mathbf{E}\mathbf{a} - \mathbf{1}_{10}). \end{aligned} \quad (17)$$

The logistic term is convex in $\mathbf{a}$. Because $\alpha > 0$ and $\mathbf{E}$ has full column rank, the ridge Hessian $(2\alpha/10)\mathbf{E}^\top \mathbf{E}$ is positive definite. The complete criterion is therefore strictly convex over the convex box constraint and has a unique minimiser.

For target season Y , let $\mathcal{C}_Y$ be the eligible completed prior-season window. If its distinct game dates are $t_1 < \dots < t_m$, the inner partitions are

$$\mathcal{T}_Y = \{g \in \mathcal{C}_Y : \text{date}(g) \in \{t_1, \dots, t_{\lfloor 0.70m \rfloor}\}\}, \quad \mathcal{V}_Y = \mathcal{C}_Y \setminus \mathcal{T}_Y. \quad (18)$$

The candidate hyperparameter set applied to each target season was

$$\begin{aligned} \mathcal{H} &= \{1, 2, 3, 5, 10\} \\ &\quad \times \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1, 3, 10\}, \end{aligned} \quad (19)$$

where the first coordinate is τ and the second is α . Let $\varepsilon = 10^{-9}$ denote the numerical zero tolerance. For each candidate, define the number of strict inner-holdout concordances and its mean inner-holdout loss by

$$H_Y(\tau, \alpha) = \sum_{g \in \mathcal{V}_Y} \mathbf{1}\{z_g(\hat{\mathbf{a}}_{\mathcal{T}_Y}(\tau, \alpha)) > \varepsilon\}, \quad (20)$$

$$\bar{\ell}_{Y, \mathcal{V}}(\tau, \alpha) = \frac{1}{|\mathcal{V}_Y|} \sum_{g \in \mathcal{V}_Y} \ell_\tau[z_g(\hat{\mathbf{a}}_{\mathcal{T}_Y}(\tau, \alpha))]. \quad (21)$$

The selection hierarchy is defined by

$$(\hat{\tau}_Y, \hat{\alpha}_Y) = \arg \max_{(\tau, \alpha) \in \mathcal{H}} (H_Y(\tau, \alpha), \alpha, -\tau, -\bar{\ell}_{Y, \mathcal{V}}(\tau, \alpha)). \quad (22)$$

It first maximises strict inner-holdout concordances. Because many candidates had identical integer counts, the deterministic tie-breaking rule then prefers stronger regularisation, a smaller transition width and finally a smaller smooth holdout loss; this

prioritises shrinkage toward official weights among count-equivalent candidates rather than treating small loss differences as primary evidence. Selection is followed by a complete-window refit,

$$\hat{\mathbf{a}}_Y = \hat{\mathbf{a}}_{\mathcal{C}_Y}(\hat{\tau}_Y, \hat{\alpha}_Y), \quad (23)$$

so no target-season observation enters hyperparameter selection or the raw coefficient estimate.

After hyperparameter selection, the chosen specification is refitted on the complete eligible calibration window. The numerical optimisation implements that bounded, strictly convex criterion as follows.

Raw magnitudes minimised the bounded criterion in Equations 15 and 16; the ridge anchor stabilises correlated weights³². Its divisor and norm refer to the ten scored fields, not the eight free groups: each member of a restricted pair appears once because it remains a separate credited or penalised ledger field. Points and point margins were excluded, and winner identity entered only through row orientation. A bounded limited-memory quasi-Newton method³³ used the analytic gradient in Equation 17 and initialisation at $\mathbf{1}_8$. Solutions within 10^{-7} of a raw bound were classified as boundary-active. Convergence was assessed by finite-difference gradient comparison and refitting from lower-bound, unit and upper-bound initialisations. The strictly positive ridge grid maintained the stated strict convexity and excluded an unregularised candidate from selection.

RMS scale matching and final RM-PIR construction

The bounded fit determines a relative nonpoint direction; RMS matching is applied only afterward to provide an interpretive scale anchor to official PIR. It matches the calibration-sample RMS of the fitted winner–loser nonpoint contrasts to the corresponding official RMS. Because the factor is positive, this post-fit step cannot change which side of zero a fitted nonpoint contrast occupies and therefore does not determine nonpoint ordering.

Let the official and fitted raw calibration contrasts be

$$o_g = \mathbf{x}_g^T \mathbf{1}_{10}, \quad q_{g,Y} = \mathbf{x}_g^T \mathbf{E} \hat{\mathbf{a}}_Y = z_g(\hat{\mathbf{a}}_Y). \quad (24)$$

For a sequence $v = \{v_g : g \in \mathcal{C}_Y\}$, define

$$\text{RMS}_{\mathcal{C}_Y}(v) = \left[\frac{1}{|\mathcal{C}_Y|} \sum_{g \in \mathcal{C}_Y} v_g^2 \right]^{1/2}. \quad (25)$$

RM-PIR imposes $\text{RMS}_{\mathcal{C}_Y}(\gamma q_Y) = \text{RMS}_{\mathcal{C}_Y}(o)$. Solving this equality for the positive common factor gives

$$\gamma_Y = \frac{\text{RMS}_{\mathcal{C}_Y}(o)}{\text{RMS}_{\mathcal{C}_Y}(q_Y)} = \sqrt{\frac{\sum_{g \in \mathcal{C}_Y} o_g^2}{\sum_{g \in \mathcal{C}_Y} q_{g,Y}^2}}. \quad (26)$$

Provided $\text{RMS}_{\mathcal{C}_Y}(q_Y) > 0$, substitution of Equation 26 yields the defining identity

$$\frac{1}{|\mathcal{C}_Y|} \sum_{g \in \mathcal{C}_Y} [\gamma_Y q_{g,Y}]^2 = \frac{1}{|\mathcal{C}_Y|} \sum_{g \in \mathcal{C}_Y} o_g^2. \quad (27)$$

This is equality of uncentred second moments, equivalently RMS values. For any sequence $v_1, \dots, v_n$,

$$\text{RMS}(v)^2 = \frac{1}{n} \sum_{g=1}^n (v_g - \bar{v})^2 + \bar{v}^2, \quad (28)$$

so equal RMS values do not imply equal centred variances when the means differ. Nor does Equation 27 preserve component distributions, player-score variance, team-score variance or the complete-score distribution. The normalisation is a scale-comparability condition applied after estimation; it is not a safeguard against a points-only fit because points never enter Equation 16.

The signed coefficient vector locked for target season Y is

$$\mathbf{c}_Y = \gamma_Y \text{diag}(\mathbf{s}) \mathbf{E} \hat{\mathbf{a}}_Y, \quad c_{j,Y} = s_j \gamma_Y (\mathbf{E} \hat{\mathbf{a}}_Y)_j. \quad (29)$$

The raw and final feasible envelopes are related by

$$0.5 \leq (\mathbf{E} \hat{\mathbf{a}}_Y)_j \leq 1.5 \implies 0.5 \gamma_Y \leq |c_{j,Y}| \leq 1.5 \gamma_Y. \quad (30)$$

Thus the box constraint regulates pre-rescaling relative magnitudes; it does not impose $[0.5, 1.5]$ on the final coefficients. For any official player or team row, the target-season RM-PIR score is

$$\begin{aligned}\text{RM-PIR}_{ug}^{(Y)} &= \text{PTS}_{ug} + \mathbf{X}_{ug}^T \mathbf{c}_Y \\ &= \text{PTS}_{ug} + \gamma_Y (\mathbf{s} \odot \mathbf{X}_{ug})^T \mathbf{E} \hat{\mathbf{a}}_Y.\end{aligned}\tag{31}$$

Points therefore retain coefficient one, while all non-PTS fields receive the signed locked coefficients in Equation 29. The official and RM-PIR non-PTS winner–loser contrasts are

$$\begin{aligned}\Delta_g^{\text{PIR,nonPTS}} &= \mathbf{x}_g^T \mathbf{1}_{10} = o_g, \\ \Delta_{g,Y}^{\text{RM,nonPTS}} &= \mathbf{d}_g^T \mathbf{c}_Y = \gamma_Y \mathbf{x}_g^T \mathbf{E} \hat{\mathbf{a}}_Y = \gamma_Y z_g(\hat{\mathbf{a}}_Y).\end{aligned}\tag{32}$$

Because $\gamma_Y > 0$, raw and scaled fitted non-PTS contrasts have the same exact zero boundary:

$$\Delta_{g,Y}^{\text{RM,nonPTS}} > 0 \iff z_g(\hat{\mathbf{a}}_Y) > 0.\tag{33}$$

Taking the winner–loser difference of Equation 31 gives

$$\begin{aligned}\Delta_{g,Y}^{\text{RM}} &= \Delta_g^{\text{PTS}} + \mathbf{d}_g^T \mathbf{c}_Y \\ &= \Delta_g^{\text{PTS}} + \gamma_Y z_g(\hat{\mathbf{a}}_Y).\end{aligned}\tag{34}$$

Consequently, complete-score discordance at numerical tolerance ε occurs exactly when

$$\Delta_{g,Y}^{\text{RM}} \leq \varepsilon \iff z_g(\hat{\mathbf{a}}_Y) \leq \frac{\varepsilon - \Delta_g^{\text{PTS}}}{\gamma_Y}.\tag{35}$$

This boundary differs from the fitted non-PTS boundary. The complete RM-PIR contrast—the primary operational outcome—is assembled by restoring the fixed point-margin contribution after nonpoint fitting; the nonpoint contrast remains the mechanism check.

RMS matching consequently serves only as a scale-comparability condition. It does not prevent a points-only fit, because points are absent from the estimator; it does not equalise centred variance, player-score variance, team-score variance or complete-score distributions; and it does not alter the nonpoint zero boundary. Equation 34 shows the final Point-A-to-Point-B connection: the complete RM-PIR winner–loser contrast is the fixed positive point margin plus the scaled fitted nonpoint contrast. Equation 35 also makes explicit that the final complete-score discordance boundary is not the same as the fitted nonpoint boundary.

Season-locked rolling-origin evaluation

This was a retrospective, non-preregistered study of official EuroLeague box scores. The model class and analytical choices were developed after historical data had been examined, so the target-season analyses are conditional retrospective backtests rather than prospective validation. Team rows were used for coefficient fitting and outcome ordering; player rows were used for exact ledger decomposition, season summaries and secondary descriptive associations and were not used to estimate the coefficients or to infer causal player impact.

For target season Y , the eligible calibration set $\mathcal{C}_Y$ comprised at most the three most recent completed seasons: 2020–21 for 2021–22; 2020–21 and 2021–22 for 2022–23; 2020–21 through 2022–23 for 2023–24; 2021–22 through 2023–24 for 2024–25; and 2022–23 through 2024–25 for 2025–26. No target-season observation entered selection, refitting or scaling.

Within each $\mathcal{C}_Y$, distinct game dates were ordered $t_1 < \dots < t_m$; games sharing a date were ordered by game identifier. Equation 18 gives the chronological 70/30 partition, Equation 19 gives the complete candidate grid, and Equation 22 formalises the selection and tie-breaking rule. The chosen specification was refitted on all of $\mathcal{C}_Y$ as in Equation 23, after which γ_Y was computed using Equation 26. Coefficients in Equation 29 were locked for every stage of target season Y , following established principles of out-of-time evaluation, dependency-aware sports-analytics evaluation and adaptation to temporal change^{34–36}. Points retained coefficient one. Raw and final bounds obey Equation 30. Specifications for which the denominator in Equation 26 was at or below 10^{-12} were defined as inadmissible; no estimated denominator met this condition.

The deployment sequence is therefore chronological at every target: form the eligible prior-season window, select (τ, α) on the chronological 70/30 inner split, refit on the complete window, compute the RMS factor in that same window, lock the signed coefficients, and only then evaluate the target season. No target-season observation enters hyperparameter selection, raw fitting or RMS scale matching.

Data sources, study population and historical comparability

The central method above is applied to the following official data and study population.

Official game reports, team statistics, player box scores and game metadata were obtained from Euroleague Basketball's public statistics service and Game Center^{31,37}. We included every completed game with official statistics from 2020–21 through 2025–26, covering the regular season, play-in, playoffs and Final Four. Unplayed fixtures were excluded, postponed fixtures were assigned to their actual playing dates and overtime games retained their complete official totals. The study comprised 2,018 games and 4,036 team-game rows; season counts were 328, 299, 328, 331, 330 and 402 games, respectively. Game Center play-by-play records were used only for the secondary 2025–26 player-exposure diagnostic described below.

The completed-game population retained 68 games from the 2021–22 regular season involving CSKA Moscow, UNICS Kazan or Zenit St Petersburg. Euroleague Basketball subsequently withdrew those clubs and removed all regular-season results against them from the calculation of the official standings⁵⁵. Retention is consistent with the estimand defined here, which concerns the concordance of completed-game statistical ledgers rather than reconstruction of the final standings. A sensitivity analysis removed all 68 games and repeated chronological selection, complete-window refitting and RMS matching wherever the affected season entered a target or calibration window.

Integrity checks required exactly two nonduplicate team rows per season–game key, unique season–game–side team-row keys, a non-tied final score, complete required fields, nonnegative attempts-minus-makes values and exact reconstruction of published official PIR. Total rebounds (TR) were read from the official total-rebounds field; only missed field goals (MFG) and missed free throws (MFT) were derived, as attempts minus makes. No event count was altered to force agreement. Technical fouls were not available as a separate field, so FC denotes the official aggregate foul-committed count and any team-only allocation is retained through the component residual. A technical-foul-specific sensitivity analysis was therefore not possible across the complete study period.

Team rows were used for fitting and outcome ordering. The 2025–26 player table includes each official player identifier with at least one positive-minute appearance across all stages, aggregating transferred players across clubs. The primary team table uses the regular season, during which all 20 clubs played 38 games; the complementary all-stage table reports cumulative and per-game values because postseason exposure differs.

The current Statistics Criteria Manual states that its criteria apply from 2024–25¹⁷. Archived 2021–22 and 2022–23 EuroLeague bylaws document a Unified Scorers Manual, scorer assessment and post-game correction procedures^{38,39}. The contemporary FIBA Statisticians' Manual documents the wider international statistical framework⁴⁰, but it does not establish which precise manual edition or field definitions governed earlier EuroLeague seasons. Season-specific Euroleague Basketball criteria manuals or a complete definition-change log for 2020–21 through 2023–24 were not located. These documents therefore support governed historical data collection but not field-by-field semantic invariance.

The ten PIR components were available under a common set of recorded fields and reconstructed official PIR exactly in every season. For component j , the adjacent-season distribution diagnostic was

$$D_{j,Y} = \frac{|\bar{X}_{j,Y} - \bar{X}_{j,Y-1}|}{\sqrt{(\sigma_{j,Y}^2 + \sigma_{j,Y-1}^2)/2}}, \quad (36)$$

where $\bar{X}_{j,Y}$ and $\sigma_{j,Y}$ are the team-game mean and population standard deviation in season Y . The largest observed value was 0.39. This diagnostic can identify a conspicuous distributional discontinuity but cannot prove semantic invariance. The rolling analysis consequently assumes material comparability of the recorded fields; a documented change in an event definition would require a new eligible calibration window.

Season aggregation, ranking and team-only residual accounting

Once a target-season coefficient vector is locked, the same row formula can be applied to official team and player observations. Aggregation and ranking are therefore downstream bookkeeping operations rather than additional coefficient estimation steps.

The same locked row formula is used for player and team observations. For score S , the season total and per-game mean are

$$S_{uY} = \sum_{g \in \mathcal{G}_{uY}} S_{ug}^{(Y)}, \quad \bar{S}_{uY} = \frac{S_{uY}}{G_{uY}}, \quad G_{uY} = |\mathcal{G}_{uY}|. \quad (37)$$

The player table uses all-stage season totals. The primary team table uses the regular-season universe; because every team has the same positive $G_{tY} = 38$, total and per-game ordering are identical. The complementary all-stage team table uses cumulative totals across unequal games played, so its cumulative ordering need not coincide with its per-game ordering. Competition rank and the reported rank change are

$$r_{u,S} = 1 + \sum_{v \neq u} \mathbf{1}\{S_{vY} > S_{uY}\}, \quad \Delta r_u = r_{u,\text{PIR}} - r_{u,\text{RM}}, \quad (38)$$

which assigns tied unrounded totals their minimum rank and makes a positive Δr_u an upward RM-PIR movement.

Official team rows need not equal the sum of their player rows. For team t , game g and component j , define

$$r_{tgj} = X_{tgj} - \sum_{i \in t} X_{igj}, \quad r_{tg,PTS} = PTS_{tg} - \sum_{i \in t} PTS_{ig}, \quad (39)$$

and the coefficient-weighted team-only residual

$$R_{tg,Y}^{RM} = r_{tg,PTS} + \sum_{j=1}^{10} c_{j,Y} r_{tgj}. \quad (40)$$

Substitution into Equation 31 yields

$$\begin{aligned} \text{RM-PIR}_{tg}^{(Y)} &= PTS_{tg} + \mathbf{X}_{tg}^T \mathbf{c}_Y \\ &= \sum_{i \in t} (PTS_{ig} + \mathbf{X}_{ig}^T \mathbf{c}_Y) + R_{tg,Y}^{RM} \\ &= \sum_{i \in t} \text{RM-PIR}_{ig}^{(Y)} + R_{tg,Y}^{RM}. \end{aligned} \quad (41)$$

The residual is therefore an exact component-wise accounting term rather than an assumed count of team rebounds or turnovers. This identity establishes algebraic decomposition across official player rows and explicit team-only events. It does not identify causal contributions, adjust for role or lineup context, or measure unrecorded actions such as screening, spacing, shot deterrence and defensive positioning.

Outcome classification and paired uncertainty

The primary and mechanism outcomes use the same winner–loser orientation established above. The following definitions specify the reported W/T/L partition, discordance rates, paired changes and uncertainty summaries.

Using the same numerical tolerance $\varepsilon = 10^{-9}$, the reported winner-higher, tie and loser-higher classification was

$$K_{S,g} = \begin{cases} \text{W}, & \Delta_{S,g} > \varepsilon, \\ \text{T}, & |\Delta_{S,g}| \leq \varepsilon, \\ \text{L}, & \Delta_{S,g} < -\varepsilon. \end{cases} \quad (42)$$

Ties and reversals were both classified as discordant under the strict winner-higher rule. Define the paired discordance indicator and target-season discordance rate as

$$I_{S,g}^{\text{disc}} = \mathbf{1}\{\Delta_{S,g} \leq \varepsilon\}, \quad r_{S,Y}^{\text{disc}} = \frac{100}{n_Y} \sum_{g=1}^{n_Y} I_{S,g}^{\text{disc}}. \quad (43)$$

The primary operational estimand was complete-score ordering in Equation 34. Nonpoint ordering was the mechanism estimand because it excludes the point margin that defines the winner. The signed paired effect reported in percentage points was

$$\hat{\delta}_Y = \frac{100}{n_Y} \sum_{g=1}^{n_Y} (I_{\text{RM},g}^{\text{disc}} - I_{\text{PIR},g}^{\text{disc}}) = r_{\text{RM},Y}^{\text{disc}} - r_{\text{PIR},Y}^{\text{disc}}. \quad (44)$$

For m discordances among n games, let $\hat{p} = m/n$ and $z_{0.975} = 1.9599639845$. Each raw discordance proportion received the descriptive Wilson interval⁴¹

$$p_{\pm} = \frac{\hat{p} + z_{0.975}^2/(2n) \pm z_{0.975} \sqrt{\hat{p}(1-\hat{p})/n + z_{0.975}^2/(4n^2)}}{1 + z_{0.975}^2/n}. \quad (45)$$

The paired transition counts relevant to improvement were

$$n_{\text{corr},Y} = \sum_g I_{\text{PIR},g}^{\text{disc}} (1 - I_{\text{RM},g}^{\text{disc}}), \quad n_{\text{new},Y} = \sum_g (1 - I_{\text{PIR},g}^{\text{disc}}) I_{\text{RM},g}^{\text{disc}}, \quad (46)$$

with the other two cells defined by both indicators being zero or both being one. Conditional on $M_Y = n_{\text{corr},Y} + n_{\text{new},Y}$, the reported two-sided exact McNemar value was computed from the Binomial($M_Y, 1/2$) distribution^{42,43}:

$$P_{\text{McN},Y} = \min \left\{ 1, 2 \sum_{k=0}^{\min(n_{\text{corr},Y}, n_{\text{new},Y})} \binom{M_Y}{k} 2^{-M_Y} \right\}. \quad (47)$$

Because the exact conditional procedure can be conservative, these values were interpreted only as exploratory secondary summaries. Like the paired-game bootstrap, this calculation treats games as paired sampling units and does not account for recurring club membership; the club-membership resampling below provides the corresponding dependence check.

Paired-game bootstrap intervals retained the two discordance indicators from the same resampled game. If J_{bk} is the k th whole-game index drawn with replacement in replicate b , then^{44,45}

$$\hat{\delta}_Y^{*(b)} = \frac{100}{n_Y} \sum_{k=1}^{n_Y} \left(I_{\text{RM},J_{bk}}^{\text{disc}} - I_{\text{PIR},J_{bk}}^{\text{disc}} \right). \quad (48)$$

The 20,000-replicate interval comprised the empirical 0.025 and 0.975 quantiles of $\hat{\delta}_Y^{*(b)}$. These game-level intervals do not account for repeated club membership. For that dependence check, sampling clubs with replacement produced a multiplicity for every club, and each target game was weighted by the product of its two clubs' multiplicities. A separate 1,000-replicate joint procedure resampled calibration games and refitted coefficients at the selected $(\hat{\tau}_Y, \hat{\alpha}_Y)$, then applied the same target-club weighting. These pigeonhole-style procedures were motivated by multiway resampling for non-independent membership structures⁴⁶; they are descriptive dependence checks, not claims that clubs or schedules were generated independently.

Tie sensitivity was assessed as a reporting analysis rather than a change to fitting. In addition to W/T/L counts, concordance awarded one half to exact ties; reversals were reported separately over all games; and a common equivalence band classified $|\Delta_{S,g}| \leq 0.5$ as a tie for both scores. The half-unit width is below the one-unit resolution of official integer PIR totals and therefore does not merge distinct official-score differences, while imposing a symmetric reporting tolerance on the continuous RM-PIR contrast. Symmetric relative error was

$$\text{SRE}(a,b) = \frac{200|a-b|}{|a|+|b|}, \quad (49)$$

with 0/0 defined as zero. Because SRE is unsigned and can be large when both errors are small, it was secondary to counts and signed changes.

Results

Study population and data integrity

The analysis comprised 2,018 completed EuroLeague games from 2020–21 through 2025–26, corresponding to 4,036 official team-game rows. The score components and evaluation quantities are defined in Table 1. Every game contained two team rows and a non-tied final score; all official team PIR values were reconstructed exactly; no required component was missing; and attempts-minus-makes was non-negative throughout. The adjacent-season distribution screen found no standardised component-mean shift larger than 0.39, although arithmetic and distributional continuity cannot establish that event-recording semantics were invariant.

Official PIR baseline under the study criterion

Before applying RM-PIR, the official ledger establishes the reference point under the author-selected winner–loser concordance criterion. Across the five rolling-origin target seasons, official complete PIR produced 150 discordances and official nonpoint PIR produced 293. In the latest 2025–26 season, official complete PIR classified 375 games as winner-higher, 3 as tied and 24 as loser-higher, so 27 of 402 games were discordant under the strict rule. The corresponding official nonpoint ledger classified 334 games as winner-higher, 14 as tied and 54 as loser-higher, so 68 games were discordant. These counts describe agreement between the completed-game team ledger and the observed winner; they are not an objective measure of player value.

RM-PIR complete-score concordance in 2025–26

For locked target-season coefficients, Equation 34 decomposes the complete winner–loser difference into the fixed point margin and the recalibrated nonpoint contrast. The exact discordance condition is given in Equation 35. Its boundary differs from the fitted condition $z_g > 0$: the complete score is the primary operational outcome, but it is not the direct optimisation target.

In 2025–26, official complete PIR produced 375 winner-higher, 3 tied and 24 loser-higher games; RM-PIR produced 381, 0 and 21. Discordances decreased from 27 to 21 (−1.49 percentage points): 9 games were reclassified into concordance and 3

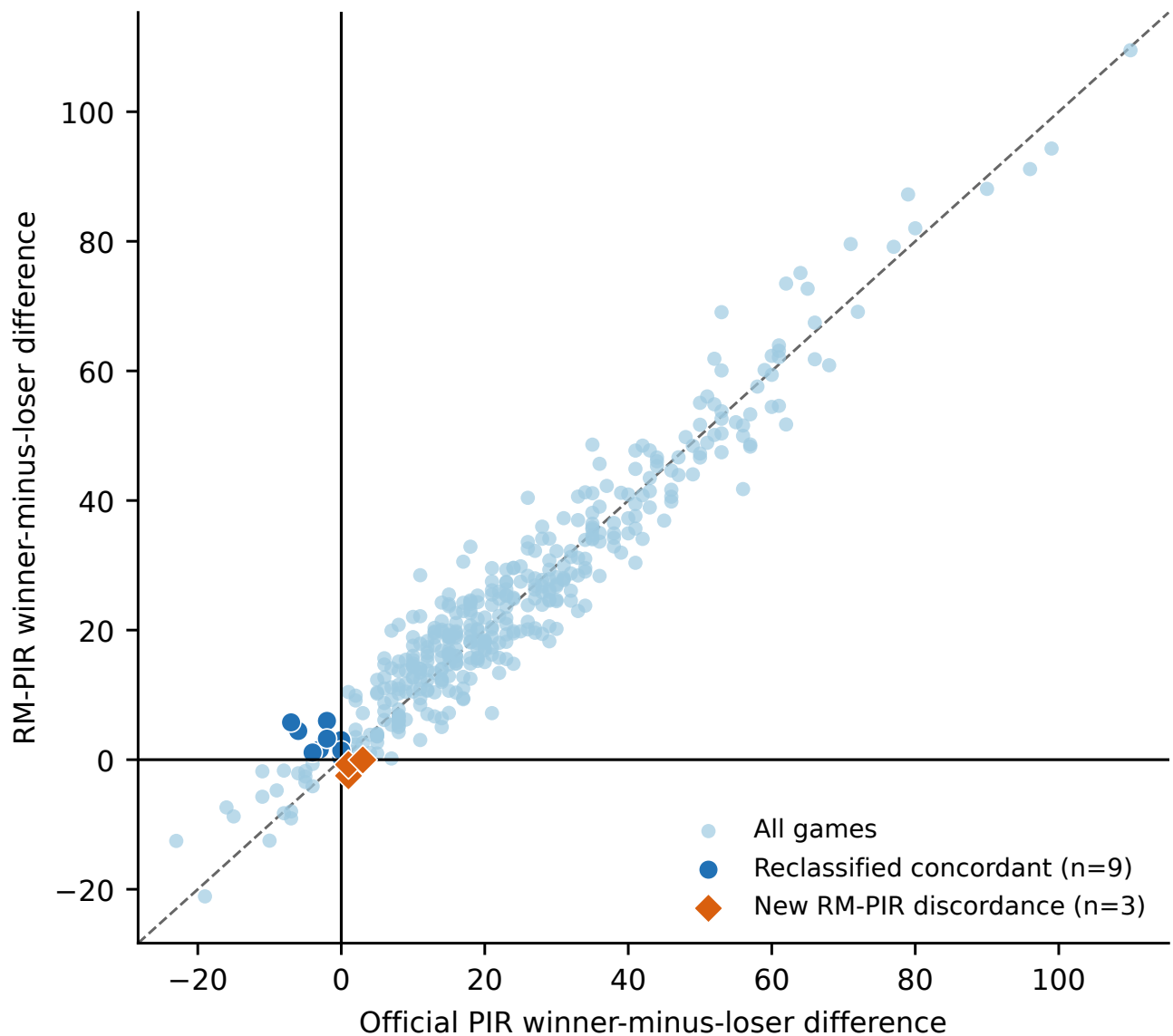


Figure 3. Official PIR and RM-PIR winner-minus-loser differences in all 402 games in 2025–26. Light-blue points show all games; dark-blue points identify the 9 official discordances reclassified as concordant by RM-PIR; orange diamonds identify the 3 newly introduced RM-PIR discordances. RM-PIR uses the coefficient vector selected and fitted from 2022–23 through 2024–25. Black axes mark the strict boundary; the dashed diagonal denotes equal differences.

became newly discordant. Of the 9 gains, 3 began as official ties and 6 as reversals. Half-credit-for-ties concordance was 93.66% for official PIR and 94.78% for RM-PIR; reversal rates over all games were 5.97% and 5.22%. Under the common 0.5-unit band, W/T/L counts were 375/3/24 and 378/4/20, reducing the strict net gain from six to three games. The club-membership interval $[-4.66, 1.49]$ and joint calibration-refit/target-club interval $[-4.55, 1.62]$ both included zero. The exact McNemar value was $P = 0.146$ and is treated as exploratory. Figure 3 shows every complete-score difference and the 372/3/9/18 paired transition table.

The complete-score result is the primary operational comparison specified in the Methods. The reclassified games therefore show how the deployed additive ledger changed under the locked recalibration; they do not imply that every changed game represents an improvement in causal basketball value.

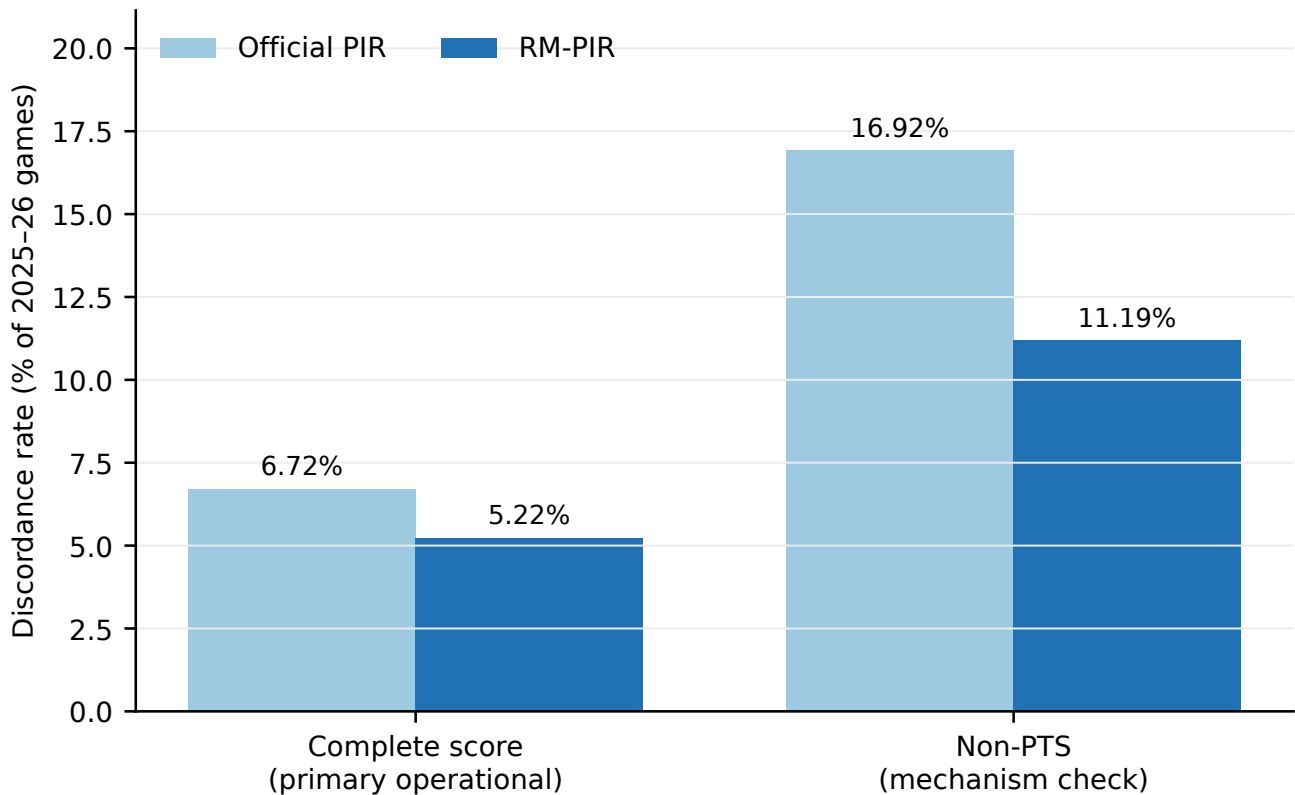


Figure 4. 2025–26 discordance rates for the primary operational complete score and the nonpoint mechanism check. Bars show the percentage of all 402 games in which the score tied the teams within tolerance or ranked the loser above the winner.

Nonpoint mechanism check in 2025–26

The nonpoint comparison removes the direct point margin that mechanically determines the winner and therefore evaluates the fitted relative action direction itself.

Official nonpoint PIR produced 334 winner-higher, 14 tied and 54 loser-higher classifications in 2025–26; the locked RM-PIR nonpoint component produced 357, 0 and 45. Discordances therefore decreased from 68 to 45 (–5.72 percentage points), with 28 games reclassified into concordance and 5 newly discordant. Ten of the 28 gains began as official ties and 18 as reversals. Half-credit-for-ties concordance was 84.83% for official nonpoint PIR and 88.81% for RM-PIR. Under a common 0.5-PIR-unit equivalence band, the corresponding W/T/L counts were 334/14/54 and 352/10/40, retaining a 4.48-percentage-point strict difference. A club-membership bootstrap interval for the strict change was $[-11.14, -0.84]$ percentage points; a joint calibration-refit and target-club interval, conditional on the selected hyperparameters, was $[-11.48, -1.90]$. Conventional game-level inference is reported only as an exploratory secondary calculation (exact McNemar $P = 6.62 \times 10^{-5}$). Figure 4 presents this mechanism check alongside the primary operational comparison.

This larger nonpoint change is the mechanism check rather than the primary deployed-score outcome. Its role is to show what happened to winner–loser ordering in the portion of the ledger that was actually fitted, before the fixed point contribution was restored.

Historical rolling-origin evidence

The focal-season comparison was also evaluated across all five season-locked historical targets. In the primary complete score, official PIR produced 150 discordances and RM-PIR produced 104, a signed change of –2.72 percentage points; 54 official discordances were reclassified into concordance and 8 games became newly discordant. In the nonpoint mechanism outcome, official PIR produced 293 discordances and RM-PIR produced 209, a signed change of –4.97 percentage points; 104 games were reclassified into concordance and 20 became newly discordant. Table 2 reports the official baseline, locked RM-PIR result and paired change for every target season. The pooled rows are descriptive rather than five independent replications: target seasons used different locked vectors, clubs recur, later calibration windows contain earlier targets, and the analytical design was developed with the historical record already inspected.

Table 2. Conditional historical rolling-origin backtest across five target seasons. W/T/L denotes winner higher, numerical tie and loser higher at tolerance 10^{-9} ; ties and reversals constitute discordance under the strict rule. Rate brackets are Wilson 95% descriptive intervals. Paired cells give, in order, both scores concordant, official PIR only concordant (a newly introduced RM-PIR discordance), RM-PIR only concordant (reclassified into concordance), and both discordant. P is the exploratory two-sided exact McNemar value. Within each calibration window, (τ, α) was selected on a chronological 70/30 inner split and then refitted on the full window before RMS matching and target-season locking. Δ disc. is RM-PIR minus official PIR discordance in percentage points; its paired-game bootstrap 95% descriptive interval is shown in brackets. SRE is unsigned and secondary.

Target	Calibration	Games	PIR W/T/L	RM W/T/L	PIR disc. [95%]	RM disc. [95%]	Paired cells	Δ disc. [95%]	P	SRE	τ	α
<i>Complete score</i>												
2021–22	2020–21	299	273/4/22	282/0/17	8.70 [6.00, 12.44]	5.69 [3.58, 8.92]	273/0/9/17	-3.01 [-5.02, -1.34]	0.004	41.9	3	0.3
2022–23	2020–21 to 2021–22	328	294/5/29	304/0/24	10.37 [7.51, 14.14]	7.32 [4.97, 10.66]	292/2/12/22	-3.05 [-5.18, -0.91]	0.013	34.5	3	0.3
2023–24	2020–21 to 2022–23	331	300/2/29	307/0/24	9.37 [6.68, 12.99]	7.25 [4.92, 10.56]	298/2/9/22	-2.11 [-4.23, -0.30]	0.065	25.5	5	0.01
2024–25	2021–22 to 2023–24	330	298/3/29	312/0/18	9.70 [6.95, 13.37]	5.45 [3.48, 8.46]	297/1/15/17	-4.24 [-6.67, -2.12]	5.19×10^{-4}	56.0	1	0.3
2025–26	2022–23 to 2024–25	402	375/3/24	381/0/21	6.72 [4.66, 9.60]	5.22 [3.44, 7.85]	372/3/9/18	-1.49 [-3.23, 0.25]	0.146	25.0	5	0.01
Pooled	–	1690	1540/17/133	1586/0/104	8.88 [7.61, 10.33]	6.15 [5.10, 7.40]	1532/8/54/96	-2.72 [-3.61, -1.83]	1.71×10^{-9}	36.2	–	–
<i>Non-PTS score</i>												
2021–22	2020–21	299	251/3/45	267/0/32	16.05 [12.33, 20.64]	10.70 [7.68, 14.72]	248/3/19/29	-5.35 [-8.36, -2.34]	8.55×10^{-4}	40.0	3	0.3
2022–23	2020–21 to 2021–22	328	272/4/52	284/0/44	17.07 [13.39, 21.52]	13.41 [10.15, 17.53]	268/4/16/40	-3.66 [-6.40, -1.22]	0.012	24.0	3	0.3
2023–24	2020–21 to 2022–23	331	274/12/45	289/0/42	17.22 [13.54, 21.66]	12.69 [9.53, 16.71]	270/4/19/38	-4.53 [-7.55, -1.81]	0.003	30.3	5	0.01
2024–25	2021–22 to 2023–24	330	266/6/58	284/0/46	19.39 [15.49, 24.00]	13.94 [10.62, 18.09]	262/4/22/42	-5.45 [-8.48, -2.73]	5.34×10^{-4}	32.7	1	0.3
2025–26	2022–23 to 2024–25	402	334/14/54	357/0/45	16.92 [13.57, 20.89]	11.19 [8.47, 14.65]	329/5/28/40	-5.72 [-8.46, -2.99]	6.62×10^{-5}	40.7	5	0.01
Pooled	–	1690	1397/39/254	1481/0/209	17.34 [15.61, 19.22]	12.37 [10.88, 14.02]	1377/20/104/189	-4.97 [-6.21, -3.73]	6.96×10^{-15}	33.5	–	–

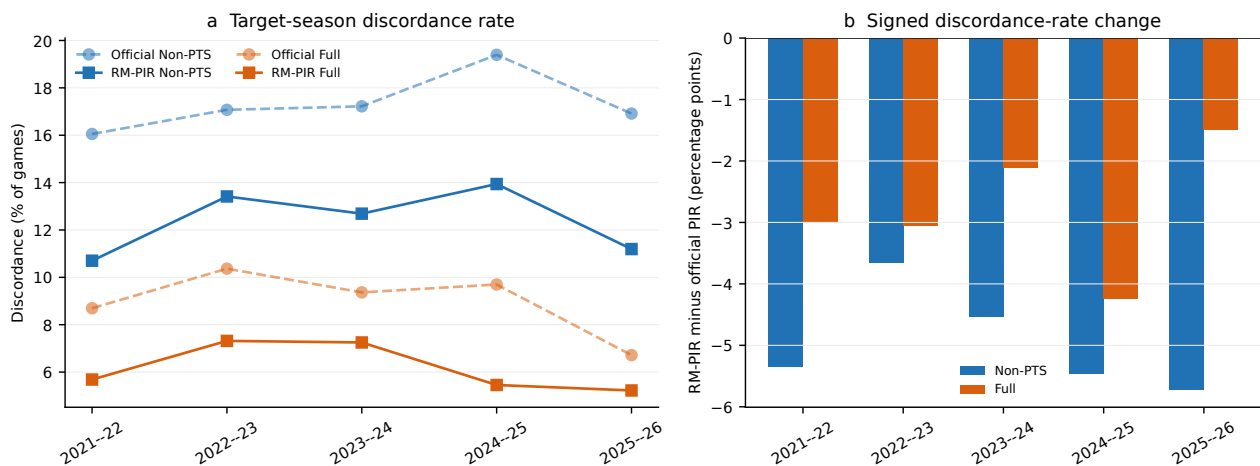


Figure 5. Five-target-season conditional historical rolling-origin backtest. (a) Official and RM-PIR discordance rates for the primary operational complete score and nonpoint mechanism outcome. (b) Signed percentage-point change, calculated as RM-PIR minus official PIR; negative values indicate fewer RM-PIR discordances. Every target-season vector excluded its target observations from inner selection, fitting and RMS matching, but the model class and analysis plan were developed after inspection of the historical period.

Figure 5 displays the same target-season sequence as discordance rates and signed RM-PIR-minus-PIR percentage-point changes, so improvements and deteriorations remain distinct rather than being compressed into the unsigned SRE summary.

Discussion

Under the author-selected outcome criterion, rolling RM-PIR produced fewer team-level discordances than official PIR in each of five historical targets. The operational complete-score effect was smaller and less robust than the nonpoint mechanism effect. In 2025–26, complete discordance decreased by 1.49 percentage points, with 9 gains and 3 losses; club-aware and joint coefficient/club intervals included zero, and a common-resolution band reduced the strict net gain. The nonpoint change was -5.72 points and remained negative under those sensitivities. This distinction is substantive: complete RM-PIR includes the point margin that defines the winner, whereas only the relative nonpoint direction is fitted.

The contribution is methodological rather than the first attempt to link basketball statistics or learned weights to team

outcomes. A close outcome-oriented team precedent is Celeux and Robert's efficiency rate, but it was estimated against season wins and evaluated through final rankings rather than locked EuroLeague completed-game ledgers⁵¹. The Total Performance Index and the later min-max PIR working paper are principally player-facing constructions^{52,53}. Econometric links from NBA player statistics to team wins also predate this study²⁷; EuroLeague work has related recorded actions to game or season outcomes^{18,21–23}; and WINSORE Revisited provides an endogenous, possession-based weighting framework evaluated with NBA and EuroLeague data and compared with PIR²⁸. Recent player and team rating frameworks pursue further estimands and structures, including hierarchical mixed-effects and principal-component ratings²⁹. RM-PIR addresses a different constrained design problem: it retains the exact official PIR fields and signs, fixes points at one, estimates bounded nonpoint magnitudes from prior seasons, restores the official calibration-window contrast RMS, locks coefficients before evaluation and preserves an explicit team-only residual. This specified combination defines the novelty claimed here; the manuscript does not claim the first data-derived EuroLeague weighting, the first outcome-oriented basketball score or a new causal measure of player value.

The rolling protocol gives each target a coefficient vector estimated only from earlier completed seasons, but it does not make the exercise prospective. The outcome criterion, action groups, bounds, RMS anchor, tuning choices and diagnostic set were developed after portions of the 2023–24 through 2025–26 record had been inspected. Earlier targets are therefore conditional backtests, not independent replications. This distinction is consistent with the broader requirement that sports-analytics evaluation respect the data-partitioning and dependency structure rather than treating model development and genuinely held-out assessment as interchangeable^{34,36}. The complete analysis specification—fields, groups, bounds, loss, tuning and tie rules, anchor and outcomes—must be frozen before an untouched 2026–27 assessment can provide prospective evidence.

The coefficient paths demonstrate transport together with active constraint dependence. The ten-field ledger contains strong pre-grouping redundancy, but the fitted eight-group design was not ill-conditioned under the reported diagnostic. Turnovers and several paired actions nevertheless reached raw bounds repeatedly, and alternative count-equivalent tuning choices produced somewhat different vectors. Similar aggregate ordering can therefore arise from different constrained coefficient vectors. The coefficients are joint parameters of a scoring direction, not causal marginal effects or context-invariant values of events. Final magnitudes may exceed the raw feasible interval after RMS matching because the positive scale factor is applied after the bounded fit; this is not a bound violation.

The player and team rankings illustrate the practical effect of reallocating action credit. They should be read as alternative summaries of the same recorded ledger. Player totals depend on exposure and role, and team-level calibration cannot establish individual construct validity merely because the formula can be evaluated on player rows. Likewise, regular-season team ranks describe average recorded production rather than standings, causal team strength or future success. Alternative player-evaluation frameworks such as regularised adjusted plus-minus and hierarchical athlete/team ratings estimate different objects under different data structures and assumptions^{29,48,49}. Contextual player assessment still requires video, lineup, possession and tracking information where available. Adjusted plus-minus, spatial defensive, possession-outcome and high-resolution tracking models address such questions with richer contextual data^{12–15}.

Several limitations qualify the results. First, exact PIR reconstruction and modest distribution shifts establish arithmetic consistency and continuity, not semantic invariance of event recording; the available official criteria manuals begin with 2024–25. The primary completed-game population also retained 68 games whose results were subsequently removed from the official 2021–22 standings. Their inclusion is consistent with the completed-game ledger estimand, and the exclusion analysis preserved every affected effect direction, but the primary sample should not be interpreted as reproducing the official standings for that season. Second, one chronological inner split was used per window and selected solutions were boundary-active. Third, the club-membership and joint resampling analyses improve on game independence but are still descriptive summaries, not a generative hierarchical model. Fourth, RMS matching preserves neither centred variance, target-season scale nor player- or full-score distributions, and it was not necessary for the sign of the nonpoint result. Fifth, strict ties favour a continuous score unless tie-equivalent sensitivities are shown. Finally, box scores omit opponent assignment, game-state leverage, off-ball actions and many defensive or spatial contributions^{4,12,15}; recorded statistics and team performance can also vary with schedule, travel and competition context⁵⁰, while the player diagnostics contain exposure and final-margin dependencies.

Within these limits, RM-PIR is a transparent, season-locked reweighting of the EuroLeague action ledger. It retains the official fields, fixes the point coefficient at one and preserves exact decomposition through explicit team-only residuals. Across five conditional historical targets, it produced more outcome-concordant team ledgers under the author-selected criterion, with larger and more consistent changes in nonpoint than complete-score ordering. These findings support prospective evaluation of a frozen specification; they do not establish that any individual's revised credit is more valid or that RM-PIR provides a causal, predictive or context-complete estimate of player impact.

Data availability

The source team and player box scores are publicly accessible through Euroleague Basketball's statistics service, and game and play-by-play records through its Game Center^{31,37}. The analysis-ready dataset is derived from these public records. Code,

configuration files and materials required to reproduce the numerical analysis will be made publicly available in a GitHub repository at publication, subject to the terms of the underlying data provider. Because the underlying provider data are not owned by the authors, redistribution of raw source records will follow the provider's terms of use.

Code availability

All code used for data processing, model fitting, sensitivity analyses, numerical validation and figure generation will be deposited in a public GitHub repository at publication. The repository will include the analysis scripts, model specification, parameter settings and instructions required to reproduce the reported results from the permitted source data.

Author contributions

Nicholas Christakis (NC): Conceptualization, theory development, methodology, validation, analysis, writing—original draft, and writing—review and editing.

Panagiotis Tirchas (PT): Simulations, software/code development, data analysis, validation, graph and figure generation, writing—original draft, and writing—review and editing.

Dimitris Drikakis (DD): Conceptualization, formal analysis, and writing—review and editing.

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Competing interests

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